

Sprint 1 Structured Notes: Code Editors and VS Code Setup

1. Current Progress

You are at the **beginning** of the Relational Databases learning path.

At this stage, you are **not learning SQL yet**. Before learning SQL and PostgreSQL, you need to prepare your coding environment.

The full learning path is:

Tools → Terminal → Bash → Bash Scripting → Git → PostgreSQL → SQL → Normalization, Security, Performance

Sprint 1 focuses on the first part: **Tools**.

In Sprint 1, you learn how to use a code editor, especially **Visual Studio Code**, usually called **VS Code**.

2. Main Goal of Sprint 1

By the end of Sprint 1, you should be able to:

Open VS Code, create a project folder, create an HTML file, run it using Live Server, and explain what a workspace is.

This is the main checkpoint:

Open an HTML file through Live Server and explain what a workspace is in one sentence.

3. What Is a Code Editor?

A **code editor** is an application used to write and edit code files.

It is like a writing tool for programmers.

Examples of code editors:

- VS Code
- Sublime Text

- Notepad++

A code editor helps you write code more easily because it can show colors, highlight mistakes, organize files, and support extensions.

Simple meaning:

A code editor is a program where developers write and edit code.

4. What Is an IDE?

An **IDE** means **Integrated Development Environment**.

An IDE is bigger than a simple code editor.

An IDE usually helps you:

- Write code
- Run code
- Debug errors
- Build applications
- Manage project files

Examples of IDEs:

- Visual Studio
- Xcode
- Android Studio

Simple meaning:

An IDE is a full software development tool that helps you write, run, test, and debug code.

5. Code Editor vs IDE

A **code editor** mainly focuses on writing and editing code.

An **IDE** gives more tools around the code, such as running, debugging, building, and testing.

Simple comparison:

Tool Type	Main Use	Examples
Code Editor	Write and edit code	VS Code, Sublime Text, Notepad++

Tool Type	Main Use	Examples
IDE	Write, run, debug, and manage applications	Visual Studio, Xcode, Android Studio

Important point:

VS Code is mainly a **code editor**, but when you add extensions, it can work like a lightweight IDE.

6. Local Editor vs Cloud Editor

There are two common types of editors:

1. **Local editors**
 2. **Cloud-based editors**
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7. What Is a Local Editor?

A **local editor** is installed and used on your own computer.

Example:

VS Code installed on your laptop or desktop computer

You can use it even when you are working with local files, terminal, Git, Bash, PostgreSQL, and SQL files.

Simple meaning:

A local editor runs on your own computer.

8. What Is a Cloud Editor?

A **cloud editor** runs inside a web browser.

You do not need to install it on your computer.

Examples:

- Replit
- GitHub Codespaces

Simple meaning:

A cloud editor lets you write and run code through the internet browser.

9. Why We Use VS Code for This Course

For this course, VS Code is recommended because you will later work with:

- HTML files
- Bash commands
- Bash scripts
- Git projects
- SQL files
- PostgreSQL practice
- README files
- Project folders

VS Code is useful because it supports many programming languages and many types of projects through extensions.

Simple meaning:

VS Code is flexible, beginner-friendly, and useful for many development tasks.

10. What Is VS Code?

VS Code stands for **Visual Studio Code**.

It is a popular code editor created by Microsoft.

It helps developers write and manage code.

VS Code can be used for many languages and project types.

Examples of things you can write in VS Code:

- HTML
- CSS
- JavaScript
- Bash scripts
- SQL
- Markdown
- Python
- README files

Simple meaning:

VS Code is the main code editor we will use to write, organize, and manage project files.

11. What Is a Workspace?

A **workspace** is the folder that you open in VS Code.

Example:

You create a folder called:

```
my-project
```

Then you open that folder in VS Code.

Now, that folder is your **workspace**.

Simple meaning:

A workspace is the project folder currently opened in VS Code.

This is very important because VS Code works around the folder you open.

If you open the wrong folder, your files may not appear in the VS Code file explorer.

12. What Is a Directory?

A **directory** means a **folder**.

In programming, people often say “directory” instead of “folder”.

Example:

```
my-project
```

This is a folder, so it is also a directory.

Simple meaning:

A directory is another word for a folder.

13. What Is a CLI?

CLI means **Command Line Interface**.

A CLI lets you control the computer using text commands.

Instead of clicking buttons, you type commands and press Enter.

Examples:

- Command Prompt
- PowerShell
- Terminal

Simple meaning:

A CLI is a text-based way to give commands to the computer.

14. Creating Your First Project Folder

To create a project folder, open your terminal or command prompt.

Then run:

```
cd ~  
mkdir my-project
```

Meaning:

```
cd ~
```

This means go to your home folder.

```
mkdir my-project
```

This means create a new folder called `my-project`.

Simple meaning:

- `cd` = change directory
- `~` = home folder

- `mkdir` = make directory / create folder
 - `my-project` = the folder name
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15. Opening the Project Folder in VS Code

After creating the folder, open it in VS Code.

You can do this in two ways.

Method 1: Using the VS Code Menu

Open VS Code.

Then choose:

File → Open Folder → my-project

Method 2: Using the Terminal

If the `code` command works on your computer, run:

```
code my-project
```

This opens the `my-project` folder in VS Code.

Simple meaning:

When you open the project folder in VS Code, that folder becomes your workspace.

16. Creating Your First File

Inside VS Code, create a new file named:

```
index.html
```

The file name should be exactly:

```
index.html
```

This is the common main file name for a simple web page.

In this file, you can write a simple HTML page.

Your first page should contain:

- One heading
- One paragraph
- Your name

Example idea:

```
Heading: My First Web Page
Paragraph: I am learning VS Code and Live Server.
Name: Your name
```

At this stage, the page does not need to look beautiful.

The main goal is to learn how to create and run a file.

17. What Is HTML?

HTML means **HyperText Markup Language**.

HTML is used to create the structure of web pages.

For Sprint 1, you do not need deep HTML knowledge.

You only need to understand that `index.html` is a web page file.

Simple meaning:

HTML is used to create the basic content and structure of a web page.

18. Why You Should Not Always Open HTML Directly

You may think you can double-click an HTML file and open it in the browser.

Sometimes that works.

But it can also cause problems.

Example problem:

- CSS may not load correctly
- Some browser features may not work properly
- File paths may behave differently

That is why we use **Live Server**.

Simple meaning:

Opening HTML directly can work, but Live Server is better because it runs your page like a real local website.

19. What Is Live Server?

Live Server is a VS Code extension.

It runs your web page on a small local server.

A local server means the website runs only on your own computer.

After installing Live Server, you can click:

Go Live

Then your browser opens the project.

The URL usually looks like this:

http://localhost:5500/

Simple meaning:

Live Server lets you run your HTML page like a website on your own computer.

20. What Does Localhost Mean?

`localhost` means **your own computer**.

When you see this URL:

```
http://localhost:5500/
```

It means the website is running on your computer, not on the public internet.

Other people cannot normally access it from the internet.

Simple meaning:

localhost means the website is running locally on your own computer.

21. What Does Port 5500 Mean?

In this URL:

```
http://localhost:5500/
```

`5500` is the port number.

A port is like a small door used by a program to communicate.

Live Server commonly uses port `5500`.

Simple meaning:

Port 5500 is the place where Live Server runs your local website.

22. How to Use Live Server

Steps:

1. Open VS Code.
2. Open your project folder.
3. Create or open `index.html`.
4. Install the Live Server extension.
5. Open `index.html`.
6. Click **Go Live**.
7. Check the browser page.
8. Look at the URL. It may be `http://localhost:5500/`.

Simple meaning:

Live Server helps you view your web page in the browser while working in VS Code.

23. Important VS Code Shortcuts

These shortcuts help you work faster.

Save

Ctrl + S

This saves your current file.

On Mac:

Cmd + S

Copy

Ctrl + C

This copies selected text.

On Mac:

Cmd + C

Paste

Ctrl + V

This pastes copied text.

On Mac:

Cmd + V

Hide or Show Sidebar

Ctrl + B

This hides or shows the left sidebar in VS Code.

On Mac:

Cmd + B

Zoom In

Ctrl + +

This makes the editor text bigger.

On Mac:

Cmd + +

Zoom Out

Ctrl + -

This makes the editor text smaller.

On Mac:

Cmd + -

Command Palette

Ctrl + Shift + P

This opens the command palette.

The command palette lets you search for commands inside VS Code.

On Mac:

Cmd + Shift + P

Simple meaning:

The command palette is like a search box for VS Code actions.

Search in Project

Ctrl + Shift + F

This searches text inside all files in your workspace.

On Mac:

Cmd + Shift + F

Find and Replace

Ctrl + Shift + H

This finds text and replaces it with different text.

On Mac:

Cmd + Shift + H

Format File

Shift + Alt + F

This formats the current file.

Formatting means arranging the code neatly.

Simple meaning:

Formatting makes your code cleaner and easier to read.

24. VS Code Extensions

An **extension** adds extra features to VS Code.

VS Code starts as a lightweight editor.

Extensions make it more powerful.

Simple meaning:

An extension is an add-on that gives VS Code extra features.

25. Recommended Beginner Extensions

For Sprint 1, install only a few useful extensions.

Do not install too many extensions at the beginning because they can confuse beginners.

Recommended extensions:

1. Live Server
 2. Code Spell Checker
 3. Error Lens
 4. Indent Rainbow
 5. Prettier
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26. Live Server Extension

Live Server runs your HTML page on a local server.

It gives you the **Go Live** button.

Use it to open your web page in the browser.

Simple meaning:

Live Server helps you preview your web page locally.

27. Code Spell Checker Extension

Code Spell Checker checks spelling mistakes in your text and code comments.

It is useful when writing:

- Comments
- README files
- Documentation
- Text content in HTML

Simple meaning:

Code Spell Checker helps you find spelling mistakes.

28. Error Lens Extension

Error Lens shows errors and warnings more clearly inside the editor.

Instead of only showing errors in a separate panel, it shows them close to the code.

Simple meaning:

Error Lens makes code errors easier to see.

29. Indent Rainbow Extension

Indent Rainbow colors indentation levels.

Indentation means the spaces at the beginning of a line.

This helps you understand code structure.

Simple meaning:

Indent Rainbow helps you see code blocks more clearly.

30. Prettier Extension

Prettier formats your code automatically.

It makes your code neat and consistent.

Simple meaning:

Prettier arranges your code properly.

31. Why You Should Not Install Too Many Extensions

Installing too many extensions can make VS Code confusing.

It may also make VS Code slower.

As a beginner, use only the extensions you need.

Simple rule:

Install fewer extensions first. Add more only when you understand why you need them.

32. Sprint 1 Practice Task

Complete these steps:

1. Install or open VS Code.
 2. Create a folder called `my-project`.
 3. Open `my-project` in VS Code.
 4. Create a file called `index.html`.
 5. Add a heading and paragraph.
 6. Install Live Server.
 7. Open the file using **Go Live**.
 8. Tell the browser URL you see.
 9. Explain what a workspace is in one sentence.
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33. Sprint 1 Checkpoint Answer Example

Question:

What is a workspace?

Good answer:

A workspace is the project folder currently opened in VS Code.

34. Common Beginner Mistakes

Mistake 1: Opening only the file instead of the folder

Wrong:

Opening only `index.html`.

Better:

Open the full `my-project` folder.

Why?

Because VS Code works better when the whole project folder is opened.

Mistake 2: Naming the file incorrectly

Wrong examples:

```
index.html.txt  
Index.html  
home.html
```

Correct:

```
index.html
```

Why?

`index.html` is the standard main page name for simple web projects.

Mistake 3: Forgetting to save the file

If you do not save, the browser may not show your latest changes.

Use:

```
Ctrl + S
```

or on Mac:

```
Cmd + S
```

Mistake 4: Opening HTML directly instead of using Live Server

Opening the file directly can work, but it may cause problems later.

Better:

Use Live Server and open the page from:

```
http://localhost:5500/
```

Mistake 5: Installing too many extensions

Too many extensions can confuse you.

Start with only the recommended beginner extensions.

35. Key Terms Summary

Term	Simple Meaning
Code editor	A program used to write and edit code
IDE	A bigger tool used to write, run, debug, and manage code
VS Code	The code editor used in this course
Local editor	An editor installed on your computer
Cloud editor	An editor used through the browser
Workspace	The project folder opened in VS Code
Directory	Another word for folder

Term	Simple Meaning
CLI	A text-based way to control the computer
HTML	Language used to create web page structure
Live Server	VS Code extension that runs a local web server
localhost	Your own computer
Extension	Add-on that gives VS Code extra features
Prettier	Tool that formats code neatly

36. Sprint 1 Final Understanding

After Sprint 1, you should understand that before learning databases, SQL, Bash, and Git, you need a clean coding environment.

VS Code is your main workspace tool.

Your project files should be organized inside a project folder.

That folder becomes your workspace when opened in VS Code.

Live Server helps you run your HTML file like a real local website.

Shortcuts and extensions help you work faster and more comfortably.

37. Final Sprint 1 Mini Assessment

Answer these questions:

1. What is a code editor?
2. What is an IDE?
3. What is the difference between a code editor and an IDE?
4. What is a local editor?
5. What is a cloud editor?
6. Why are we using VS Code?
7. What is a workspace?
8. What is a directory?
9. What command creates a folder?
10. What file should you create first in the project?
11. What is Live Server?
12. What does localhost mean?

13. What shortcut saves a file?
 14. What extension formats code?
 15. Why should beginners avoid installing too many extensions?
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38. Sprint 1 Completion Checklist

Tick these when completed:

- ☐ I opened VS Code.
 - ☐ I created a folder called `my-project`.
 - ☐ I opened `my-project` in VS Code.
 - ☐ I created `index.html`.
 - ☐ I wrote a heading and paragraph.
 - ☐ I installed Live Server.
 - ☐ I clicked Go Live.
 - ☐ I saw a browser URL like `http://localhost:5500/`.
 - ☐ I can explain what a workspace is.
 - ☐ I practiced the main VS Code shortcuts.
 - ☐ I understand the recommended beginner extensions.
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End of Sprint 1 Notes

You are ready to move to Sprint 2 only after you can complete the Sprint 1 task and explain the key terms in simple words.